

motherboard from ASUS, Gigabyte or MSI, 2 GB of DDR II RAM from Transcend or Kingston, 250 GB SATA hard drive from WD or Seagate, 17" LCD display from Viewsonic, LG, Dell etc and importantly NVIDIA 8500GT 512MB display card. This configuration along with common components shouldn't cost more than Rs 30,000.

Configuration 3: Intel Core 2 Duo E8400. NVIDIA 8200 or Intel P35 chipset motherboard from ASUS, Gigabyte or MSI, 2 GB of DDR II RAM from Transcend or Kingston, 500 GB hard drive from WD or Seagate, 19" LCD display from Viewsonic, LG, Dell etc and NVIDIA Quadro FX570 card. This configuration along with common components won't cost above Rs 40,000.

Common components include: 16X DVD optical drive from LG or Sony, VIP cabinet and 400W power supply, keyboard and mouse from Logitech.

Servers

Once computers are put on a network, sharing of files and resources ensues. How often and heavily each person relies on such shared resources determines whether there is a real need for a specialised, dedicated server. A server is a centralised computer on the network which accepts requests from clients (PCs) and serves the required data to those clients. Servers can be put to many uses—file sharing, database hosting, mail server, or as a web server. The role it plays generally determines the hardware requirements.

Most small offices often deploy a general PC as a server for storing and sharing files. While it might be okay to do so, I will recommend going in for branded entry level server. Reliability is the word that you should associate with servers. While it is possible to assemble a server using server components, I suggest you opt for branded entry-level servers. Each component right from motherboard to the chassis has to be designed with reliability in mind, which doesn't happen when assembling a server. It's better to look out for branded servers where manufacturers rigorously test every factor that affects reliability. Also, entry level servers for small 25 member team are cheaper than you'd expect.

Dell, HP and IBM are some of the prominent vendors in the server space. They offer all kinds



Dell's PowerEdge 840: An affordable server



Western Digital's 1 TB MyBook is perfect for offline backups

of servers—right from entry-level towers to high-end blade servers. Dell has the PowerEdge series, while with HP you can opt for the ProLiant ML series of servers. In IBM, you can choose the SystemX series of servers. Most servers from the above mentioned vendors are available for under Rs 50,000 with the basic configuration; you may add components later as per your needs.

Once the hardware's taken care of, a server will also require an equally competent server operating system. While Linux has proved its mettle in the server space, you will need an on-site system administrator to oversee matters. Microsoft does offer the Windows Server operating system, but for a small team of 10-25 members, Linux is a better alternative. You will need the help of a Linux expert when setting up initially, but once the system is set up properly, it will rarely go down.

Backup Solutions

Any organisation, no matter how big, should have an active data backup policy. Digital data is volatile and all it takes is one hard drive crash to completely wipe out your important information. You can have both online as well as offline data backup policies to ensure data redundancy. Most small offices use CDs / DVDs to take backup of data—I wouldn't advise it as the only solution, mainly because it's not reliable and not very pocket-friendly.

If you buy a server, implementing an online backup solution is simple and effective. There are many free utilities available on the internet which allows scheduled backup of your data on to the server. These nifty utilities allow complete backup, incremental backup and also restoring of data. SyncBack (www.2brightsparks.com) is one such backup client available free of cost (one version prior to the latest). The advantage of having an online backup is that if any client machine crashes, that person can always use some other machine and continue working on already available data.

Apart from having an online backup, you can also have an offline backup plan. There are two ways of doing this—have an external hard drive which you connect to the system only when taking backup. Once the backup is done, disconnect the hard drive—this extends its lifetime. External hard drives are getting cheaper and are now even available with Ethernet ports which improve the data transfer speeds over USB. Maxtor, Western Digital, Seagate and Freecom are some of the vendors offering excellent external hard drive products.

Second option for offline data backup is to use CDs / DVDs. If you plan to take data backup on CDs / DVDs then you may want to have

a look at an application called LANwriter (www.lanwriter.com). LANwriter is a centralised data publishing application which can be installed on the server and only authorised users can burn discs. This allows installing only one writer in the office and keeping track of the data that is burned on CDs / DVDs.

Internet, Email And Security

Even small companies today need a fast Internet connection. As a team of 10-25 members, you can approach vendors such as VSNL (Tata), Bharati, BSNL and MTNL. These vendors have extensive network throughout our country and will get you reliable Internet connections. If your usage is limited to e-mail and Internet browsing, then even a monthly pay-as-you-use scheme will do—in any case, don't settle for anything less than 512 Kbps. Hosting a proxy server is the best way to share your Internet connection. Make sure to block certain files (MP3s, EXEs and so on) if you have a pay-as-you-use scheme. Even an old PC with a Pentium 4 processor, 512 MB of RAM and new hard drive would prove good enough for hosting a proxy server.

For an e-mail solution, you can easily host your own email server, but I recommend Google Apps instead. All you have to do is register your domain and then follow the steps given by Google to activate your Google Apps account. The free standard version gives you 100 e-mail accounts each with 6 GB of space (and counting)—more than enough for any venture.

Guarding your network against viruses is critical, even if you maintain a regular backup of data. Viruses can infect applications, destroy data and spread to every machine of the network, lowering productivity and efficiency of your team. Internet, USB pen drives are some of the prominent entry points whereby machine gets infected. Install anti-virus software on every client machine on your network, including the server machine. Don't blindly install free antivirus tools like Avast or AVG—they're only free for personal use, so using them in a business environment will invite the kind of trouble you don't ever want. Kaspersky, Symantec and McAfee have products aimed at small to medium businesses, and Avast's professional version is also cost-effective and a good performer. If you have a server in place you may also go for centralised solutions where the server will run periodic scans on client systems. Apart from antivirus software, spyware



The Linksys NAS200 is good for those on a budget

busters such as Spybot and Ad-Aware can also be deployed to keep malicious spyware at bay.

Back Office

All offices need peripherals such as printers, scanners, faxes etc. While you may consider buying them separately, I suggest multi-functional devices (MFD) to replace them all. MFDs may be a bit expensive to buy, but work out cheaper in the long run. HP, Canon, Epson and Brother have an extensive lineup of products in this category. Before you buy any product, ensure easy availability of cartridges and services.

If you print a lot, then you might consider adding a heavy-duty printer along with an MFD. When you buy a printer, first calculate the average number of pages you print in a month. Although the market offers cheap laser printers, avoid the tendency to put one on each desk—it's better to opt for a heavy-duty mono laser printer with useful features such as double-sided printing, which saves paper. Some new printers also offer advanced features such as keying in a code to retrieve printout—each team member will be given a specific code



HP's DeskJet Pro series of MFDs are killer products for SMBs

which he has to key in for the printer to print otherwise the printer won't print his job; this dramatically reduces paper and toner wastage. HP's LaserJet P4515 is one such model. In case you want colour printouts, HP's Officejet Pro series of printers are definitely recommended.

I hope this guide helps you in setting up a robust IT infrastructure. If you have any specific queries you can e-mail them to me (or anyone from the team) and we'll make it a point to help you out. ☒

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